Lecture 10: Strength Theory

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Note: First (maximum) principal stress=σ_1

Second principal stress= σ_2

Third principal stress=σ_3

•$\sigma_1 \geq \sigma_2 \geq \sigma_3$

•For 2D plane stress, we denote principal stresses as σ_a and σ_b while another is 0.

Note: First (maximum) principal strain=ε_1

Second principal strain= ε_2

Third principal strain= ε_3

•$\varepsilon_1 \geq \varepsilon_2 \geq \varepsilon_3$

•All theories in this lecture are not dependent on coordinate system you are using, as long as stress state are the same.

Strength Condition for Simple Stress States

- **Tensile/compressive stress:**

$$\sigma_{\max} = \left(\frac{F_N}{A}\right)_{\max} \le [\sigma]$$

- **Torsional shear stress:**

$$\tau_{\max} = \left(\frac{T}{W_P}\right)_{\max} \le [\tau]$$

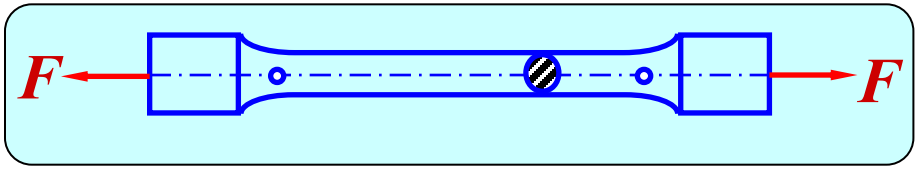
- **Bending normal stress:**

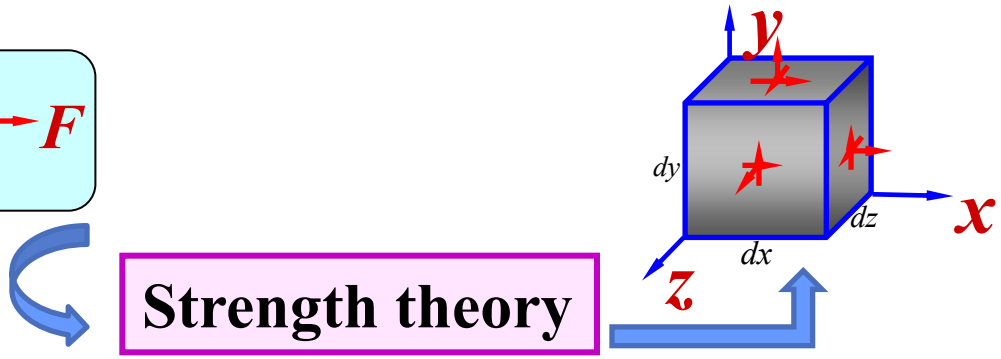
$$\sigma_{\max} = \left(\frac{M_Z}{W_z}\right)_{\max} \le [\sigma]$$

- **Bending shear stress:**

$$\tau_{max} = \left(\frac{VQ}{bI}\right)_{max} \le [\tau]$$

- **Strength condition at locations under complicated stress state:**





- **Preliminary: find the three principal stress components σ_1, σ_2 and σ_3.**

On the Damage Mechanism of Materials

- Damage (fracture) of brittle materials (concrete, glass, fiber, cast iron) may be governed by the maximum tensile stress
- Damage (yielding) of ductile materials (mild steel) may be governed by the maximum shear stress
- The goal of stress state analysis is to determine the surface/orientation on which normal/shear stress achieves the maximum values. Strength analysis can be subsequently performed in terms of the maximum normal/shear stresses.
- We assume three principal stresses in 3D cases are $\boldsymbol{\sigma_1, \sigma_2}$ **and** $\boldsymbol{\sigma_3}$
- First principal stress σ_1 is maximum and σ_3 is minimum
- For 2D plane stress, we assume principal stresses in plane are $\boldsymbol{\sigma_a}$ **and** $\boldsymbol{\sigma_b}$**,** For plane stress, there is always out-of-plane stress of 0.

Fracture Criteria for Brittle Materials

- Brittle materials fail suddenly in tensile tests. The failure condition is characterized by **the ultimate strength σ_U or ultimate strain ε_U from test results**

1. Maximum tensile stress criteria

Also called Rankine criteria or maximum principal stress criteria

Failure criteria: $\sigma_1 = \sigma_U$

Strength condition: $\sigma_1 \le \sigma_U$

- Accurate for materials primarily under tensile stress state.
- No influences from σ_2, σ_3 are taken into consideration.
- σ_1 must be tensile stress.

2. Maximum tensile strain criteria

Failure criteria: $\varepsilon_1 = \varepsilon_U$

Strength condition:

$$\varepsilon_1 = \frac{\sigma_1}{E} - \frac{\nu}{E}\left(\sigma_2 + \sigma_3\right) \le \varepsilon_U$$

$$\Rightarrow \sigma_1 - \nu\left(\sigma_2 + \sigma_3\right) \le \sigma_U$$

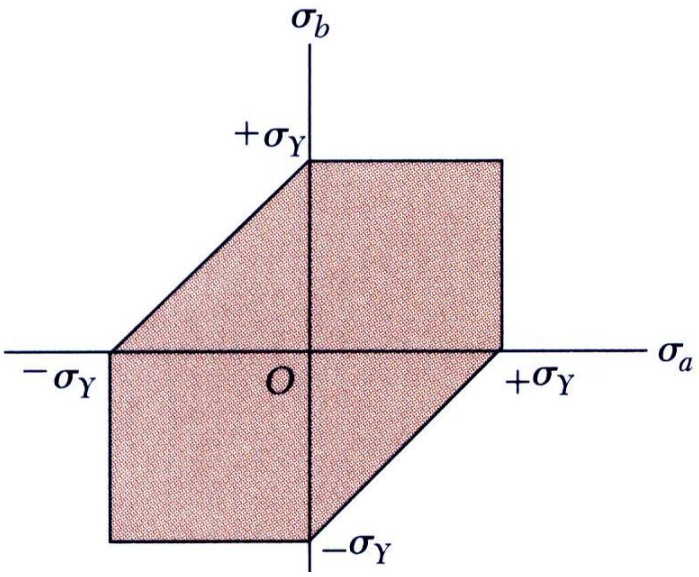
- Only applicable to linearly elastic brittle materials (until failure)
- Only accurate for few brittle materials

Yield Criteria for Ductile Materials

For concrete, even at hydrostatic stress state, the concrete material may fail under very high stress level (although this is less likely to happen in civil engineering).

For ductile material like steel, if we apply hydrostatic pressure at very high stress level (such as put a steel cube into deep see), the lattice structure does not fail (although the spacing between atoms tend to be closer). Therefore, hydrostatic stress state does not induce failure for ductile material.

Yield Criteria for Ductile Materials



Stress space plot

Horizontal and vertical axis are principal stresses in 2D

- Conservative criteria and widely adopted.
- No influences from σ_2 is taken into consideration.
- Not related to hydrostatic pressure.
- Note: σ_a, σ_b and σ_c are three principal stresses for 2D problem. We don't know which one is largest. σ_Y is uniaxial yield strength obtained from tests.

3. Maximum shear stress (Tresca) criteria

1D at yielding : $\tau_{\max} = \tau_Y = \sigma_Y / 2$

$2D\left(\sigma_a \neq 0, \sigma_b \neq 0, \sigma_c = 0\right)$:

For σ_a and σ_b with the same sign,

$$\tau_{\max} = \max\left(\frac{|\sigma_a|}{2}, \frac{|\sigma_b|}{2}\right) = \frac{\sigma_Y}{2};\ \max\left(|\sigma_a|, |\sigma_b|\right) \leq \sigma_Y$$

For σ_a and σ_b with opposite signs,

$$\tau_{\max} = \frac{|\sigma_a - \sigma_b|}{2} = \frac{\sigma_Y}{2};\ |\sigma_a - \sigma_b| \leq \sigma_Y$$

$3D\left(\sigma_1 \neq 0, \sigma_2 \neq 0, \sigma_3 \neq 0\right)$:

$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} = \frac{\sigma_Y}{2};\ \sigma_1 - \sigma_3 \leq \sigma_Y$$

Yield Criteria for Ductile Materials

4. Maximum distortion energy (von Mises) criteria σ_{eff}

von Mises stress is a non-negative scalar and also referred to as effective stress $\sigma_{eff.}$ von Mises criteria is an efficient way to evaluate the stress value in material point. It is mostly adopted for yielding failure of materials. It shows the distortion energy density at a material point. The total stress can be decomposed into hydrostatic stress vector and distortion stress vector. Von Mises assume the hydrostatic stress will not induce any damage to ductile metal, while energy related to distortion stress will induce damage to ductile metal.

von Mises stress is independent change of hydrostatic stress state (i.e, if all normal stresses increase or decrease by any constant value). In other words, suppose you drop a small metal material in deep see, the von Mises stress at that point will be 0 and no damage will happen.

Yield Criteria for Ductile Materials

4. Maximum distortion energy (von Mises) criteria σ_{eff}

4.1 The most general form of von Mises criteria for 3D (no restrictions):

$$\sigma_{\text{eff}} = \sqrt{\frac{1}{2}\left[(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2\right] + 3\left(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2\right)} \qquad \textbf{(1) remember}$$

Where σ_x σ_y and σ_z are the normal stresses in x, y, and z axis.
τ_{xy}, τ_{yz} and τ_{zx}

4.2 When the stresses are principal stresses, Eq. (1) simplified as below:

$$\sigma_{\text{eff}} = \sqrt{\frac{1}{2}\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right]} \qquad \textbf{(2) remember}$$

Where σ_1, σ_2 and σ_3 are three principal stresses.

4.3 When the stresses are plane stress, Eq. (1) simplified as below:

$$\sigma_{\text{eff}} = \sqrt{\frac{1}{2}\left[(\sigma_x - \sigma_y)^2 + (\sigma_y)^2 + (\sigma_x)^2\right] + 3\tau_{xy}^2} = \sqrt{\sigma_x^2 - \sigma_x\sigma_y + \sigma_y^2 + 3\tau_{xy}^2} \qquad \textbf{(3) remember}$$

Where σ_1, σ_2 and σ_3 are three principal stresses.

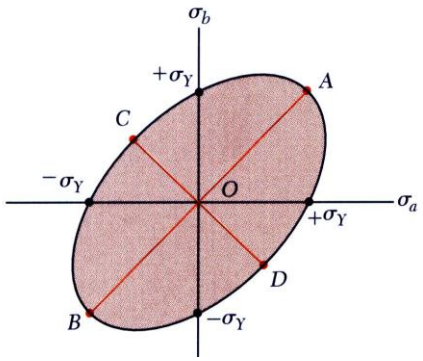
Yield Criteria for Ductile Materials

4. Maximum distortion energy (von Mises) criteria σ_{eff}

4.4 When the stresses are principal plane stress, Eq. (3) simplified as:

$$\sigma_{\text{eff}} = \sqrt{\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2} \qquad \textbf{(4)}$$

Where σ_1 and σ_2 are principal plane stresses



2D Stress space plot
Horizontal and vertical axis are principal stresses in 2D

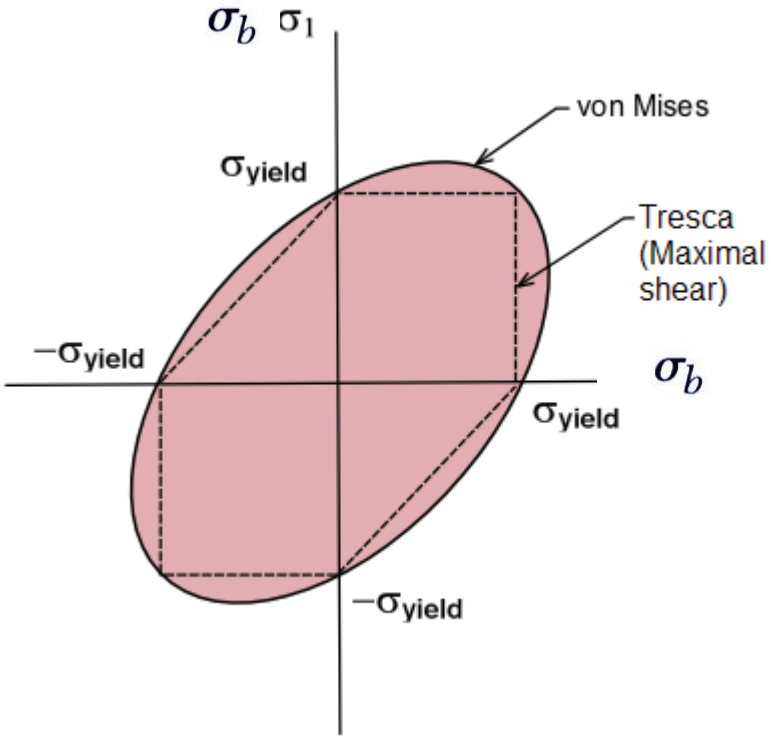
4.3 When we only have a normal stress σ_x and a shear stress τ_{xy}, and all other stress components are 0. This is the case for beams loaded under bending moment, axial force and shear force.

$$\sigma_{\text{eff}} = \sqrt{\sigma_x^2 + 3\tau_{xy}^2} \qquad \textbf{(5)}$$

Where σ_x is normal stress and stress τ_{xy} is shear stress. Need to assume all other stress components are 0 to derive Eq. (5).

Yield Criteria for Ductile Materials

Comparison between Tresca and Von Mises criterion in 2D stress space:



Suppose the yield strength is kept same, in 2D stress state figure shows Tresca criterion is slightly conservative (cover less area) compared to von Mises criterion.

Summary of Four Strength Theories

All of the four types of strength theory can be written in a universal form, in terms of an effective stress σ_r:

1. Maximum tensile stress criteria: $\sigma_1 \le [\sigma]$

2. Maximum tensile strain criteria: $\sigma_1 - \nu(\sigma_2 + \sigma_3) \le [\sigma]$

3. Maximum shear stress: $\sigma_1 - \sigma_3 \le [\sigma]$

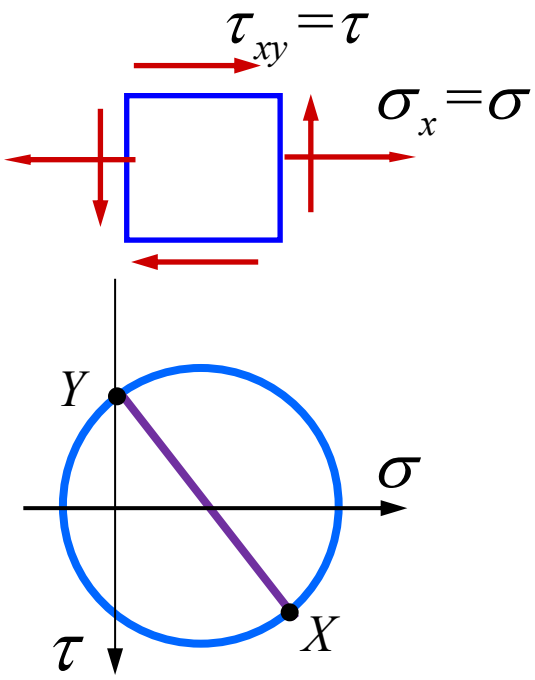
4. Maximum distortion energy criteria:

$$\sigma_{eff} = \sqrt{\frac{1}{2}\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right]} \le [\sigma]$$

Note: the limit stress has been replaced by the allowable stress.

Example 8.1 Strength criterion for 4 cases

- For the plane stress state shown, analyze the strength condition based on the four types of strength theory.



- **Solution**

$$\begin{matrix}\sigma_a \\ \sigma_b\end{matrix} = \frac{\sigma_x+\sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_{xy}^2}$$

$$\Rightarrow \begin{matrix}\sigma_a \\ \sigma_b\end{matrix} = \frac{\sigma}{2} \pm \frac{1}{2}\sqrt{\sigma^2+4\tau^2}$$

$$\sigma_1 = \frac{\sigma}{2}+\frac{1}{2}\sqrt{\sigma^2+4\tau^2},\ \sigma_2=0,\ \sigma_3=\frac{\sigma}{2}-\frac{1}{2}\sqrt{\sigma^2+4\tau^2}$$

1. Maximum tensile stress criteria:

$$\sigma_1 \le [\sigma] \Rightarrow \sigma_1 = \frac{\sigma}{2}+\frac{1}{2}\sqrt{\sigma^2+4\tau^2} \le [\sigma]$$

Example 8.1 Strength criterion for 4 cases

2. Maximum tensile strain criteria:

$$\sigma_1 - \nu(\sigma_2 + \sigma_3) \le [\sigma]$$

$$\Rightarrow \frac{(1-\nu)}{2}\sigma + \frac{(1+\nu)}{2}\sqrt{\sigma^2 + 4\tau^2} \le [\sigma]$$

3. Maximum shear stress (Tresca criterion):

$$\sigma_1 - \sigma_3 \le [\sigma] \quad \Rightarrow \sqrt{\sigma^2 + 4\tau^2} \le [\sigma]$$

4. Maximum distortion energy criteria (von Mises criterion):

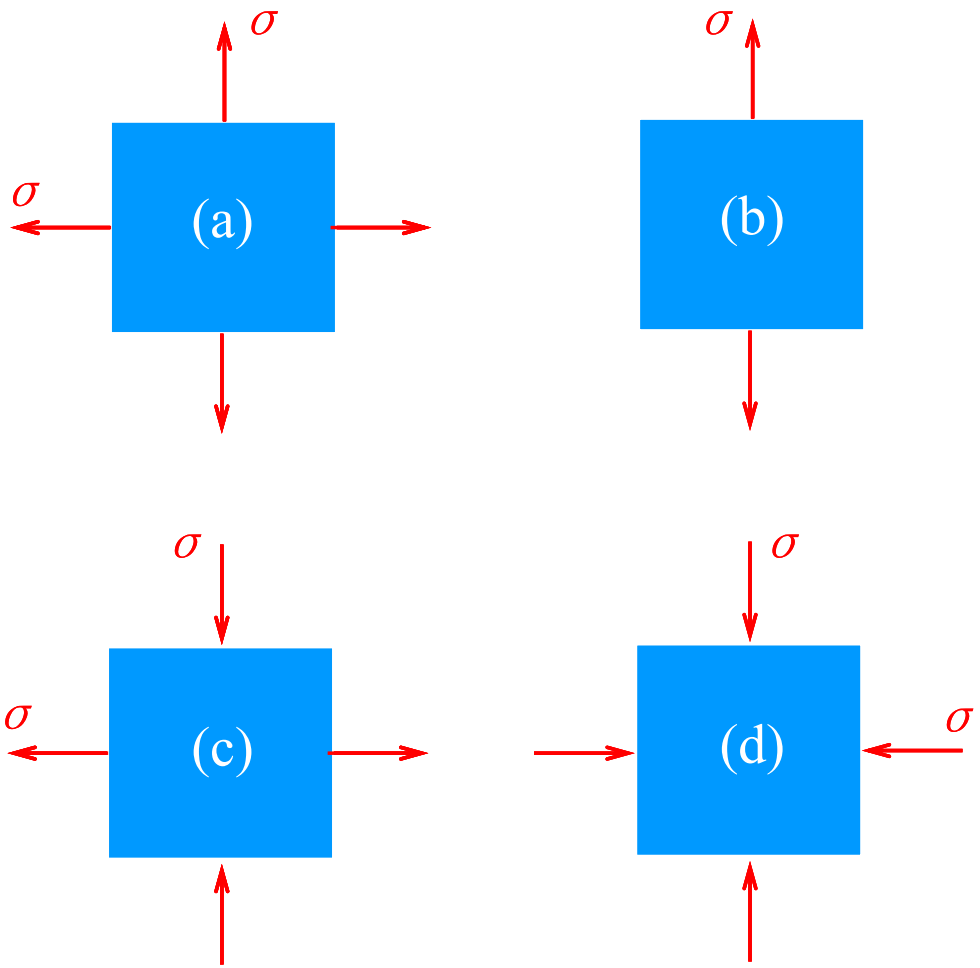
$$\sigma_{eff} = \sqrt{\frac{1}{2}\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right]} \le [\sigma]$$

$$\Rightarrow \sigma_{eff} = \sqrt{\frac{1}{2}\left[\left(\frac{\sigma}{2} + \frac{1}{2}\sqrt{\sigma^2 + 4\tau^2}\right)^2 + \left(\frac{\sigma}{2} - \frac{1}{2}\sqrt{\sigma^2 + 4\tau^2}\right)^2 + \left(\sqrt{\sigma^2 + 4\tau^2}\right)^2\right]}$$

$$\Rightarrow \sigma_{eff} = \sqrt{\sigma^2 + 3\tau^2} \le [\sigma]$$

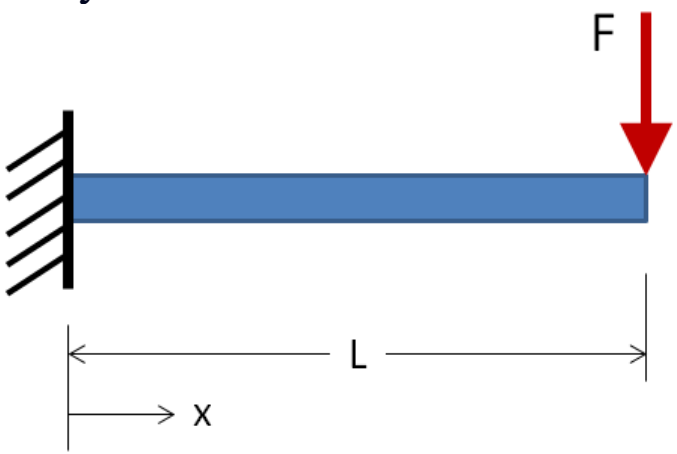
Exercise

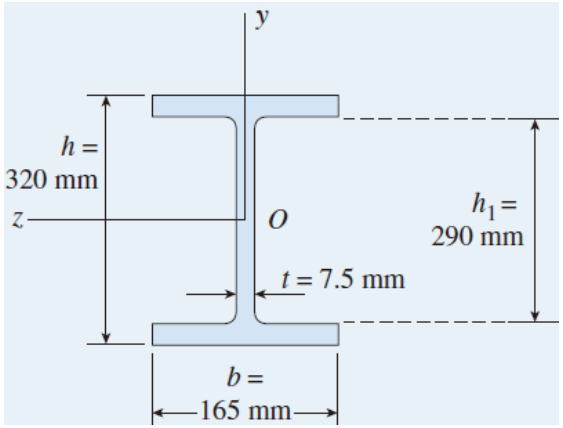
- Based on the maximum shear stress criteria, which one of the following stress states is most dangerous?



Example 8.2 Simple beam with I-shaped section

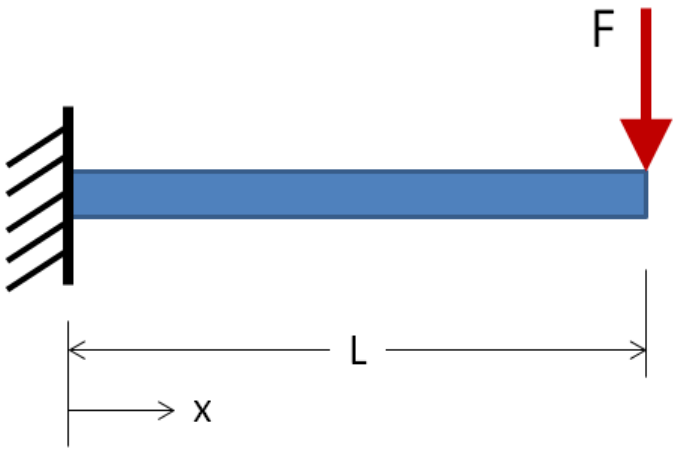
A wide-flange I-shaped beam is a cantilever beam. It has cross section as shown below and subject to vertical load of 225kN at free end. The length is L=1 m. The strength criterion is von-Mises criterion. The allowable strength is 290 MPa. Neglect self-weight of beam. Check safety of the beam.

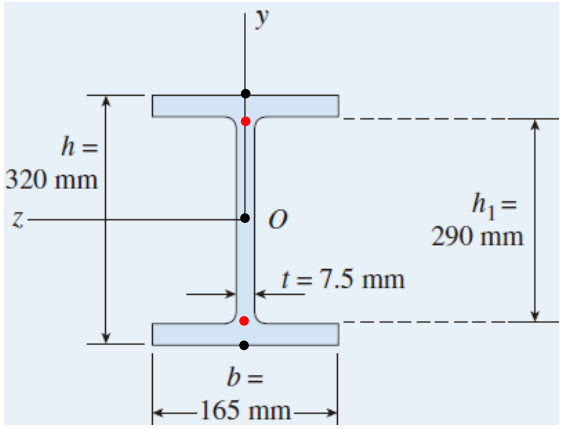




Example 8.2 Simple beam with I-shaped section

A wide-flange I-shaped beam is a cantilever beam. It has cross section as shown below and subject to vertical load of 225kN at free end. The length is L=1 m. The strength criterion is von-Mises criterion. The allowable strength is 290 MPa. Neglect self-weight of beam. Check safety of the beam.





$$I = \frac{1}{12}(bh^3 - bh_1^3 + th_1^3) = 130.45 \times 10^6\ \text{mm}^4$$

On top and bottom flange: $\sigma_{\max} = \dfrac{My_{\max}}{I} = \dfrac{225000\text{N} \times 1000mm \times 160\text{mm}}{130.45 \times 10^6\, mm^4} = 275.97MPa < 290MPa$

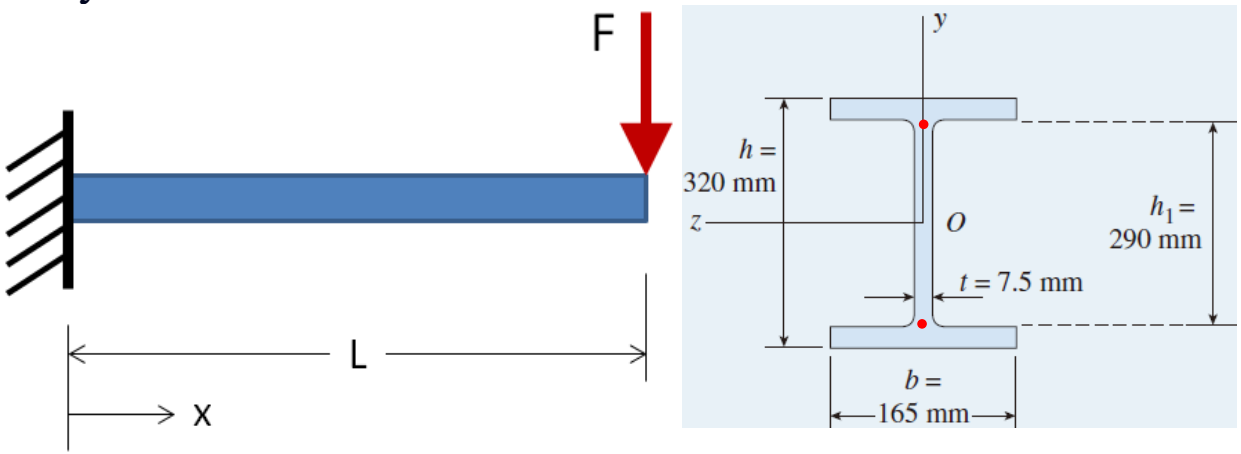
At intersection between flange and web: $\sigma = \dfrac{M \times 0.5h_1}{I} = \dfrac{225000\text{N} \times 1000mm \times 145\text{mm}}{130.45 \times 10^6\, mm^4} = 250.1MPa < 290MPa$

$$\tau = \frac{VQ}{It} = \frac{V}{8It}\left(bh^2 - bh_1^2\right) = 87MPa$$

$$\sigma_{\text{eff}} = \sqrt{\sigma^2 + 3\tau^2} = \sqrt{250.1^2 + 3 \times 87^2} = 291.99MPa > [\sigma] = 290MPa$$

Example 8.2 Simple beam with I-shaped section

A wide-flange I-shaped beam is a cantilever beam. It has cross section as shown below and subject to vertical load of 225kN at free end. The length is L=1 m. The strength criterion is von-Mises criterion. The allowable strength is 290 MPa. Neglect self-weight of beam. Check safety of the beam.

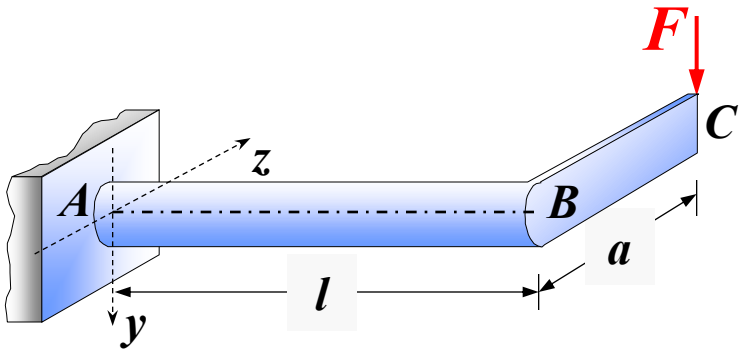


At middle height of beam, shear stress is maximum and normal stress is 0:

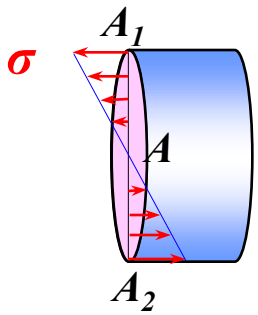
$$\tau = \frac{VQ}{It} = \frac{V}{8It}\left(bh^2 - bh_1^2 + th_1^2\right) = 105MPa \qquad \sigma_{\text{eff}} = \sqrt{\sigma^2 + 3\tau^2} = \sqrt{3\times 105^2} = 182MPa < [\sigma] = 290MPa$$

Conclusion: failure happen at the intersection between flange and web (red points noted in figure). Design is unsafe, need to change the design to meet safety requirement. Note: if axial force co-exist, need to check for both top and bottom points respectively.

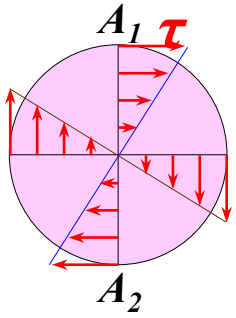
Bending & Torsion



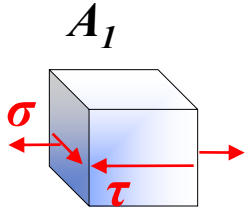
- Critical section: A

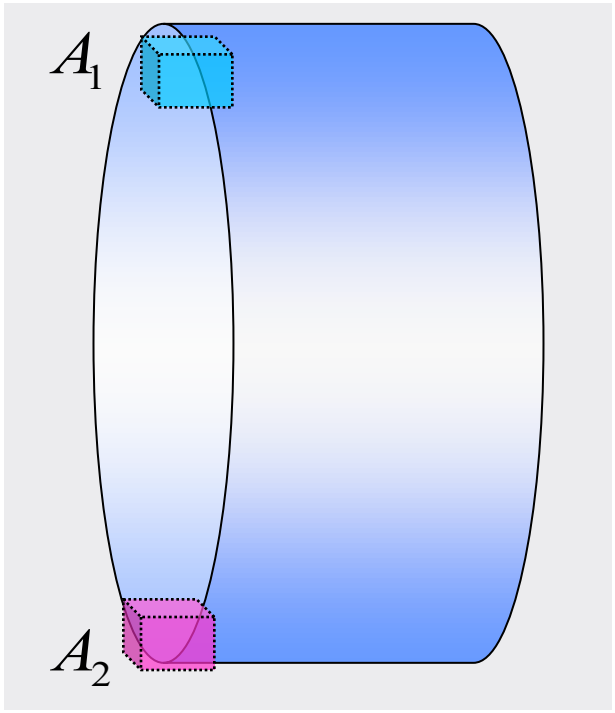


View from left



View from right





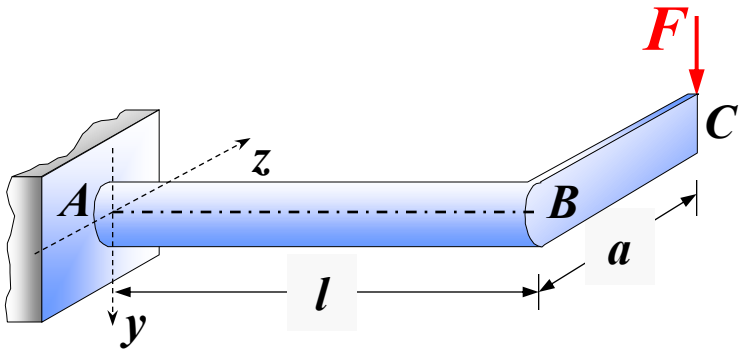
- Critical points on A: A_1 & A_2

$$\sigma = \frac{M_z}{W_z}, \quad \tau = \frac{T}{W_P}$$

$$W_z = \frac{I_z}{d/2} = \frac{1}{64}\pi d^4 / \frac{d}{2} = \frac{1}{32}\pi d^3$$

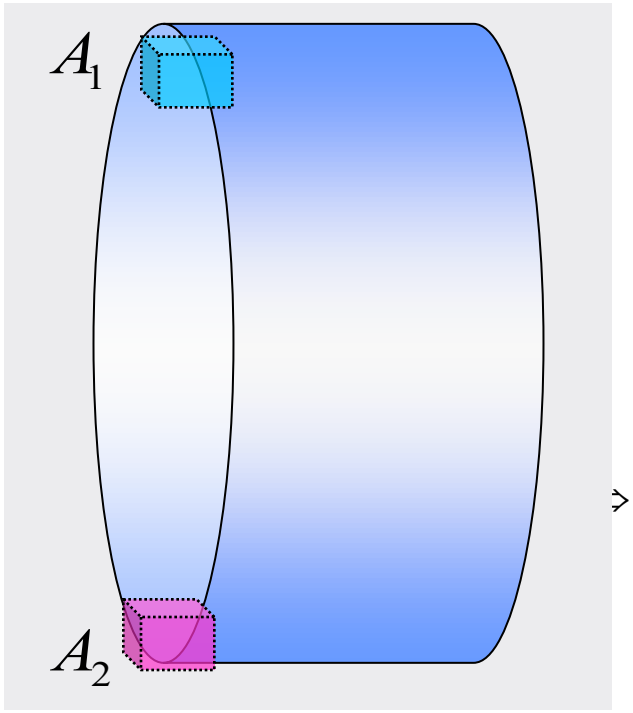
$$W_P = \frac{I_P}{d/2} = \frac{1}{32}\pi d^4 / \frac{d}{2} = \frac{1}{16}\pi d^3 = 2W_z$$

Bending & Torsion



- Critical points on A: A_1 & A_2

$$\sigma = \frac{M_z}{W_z}, \quad \tau = \frac{T}{W_P}$$



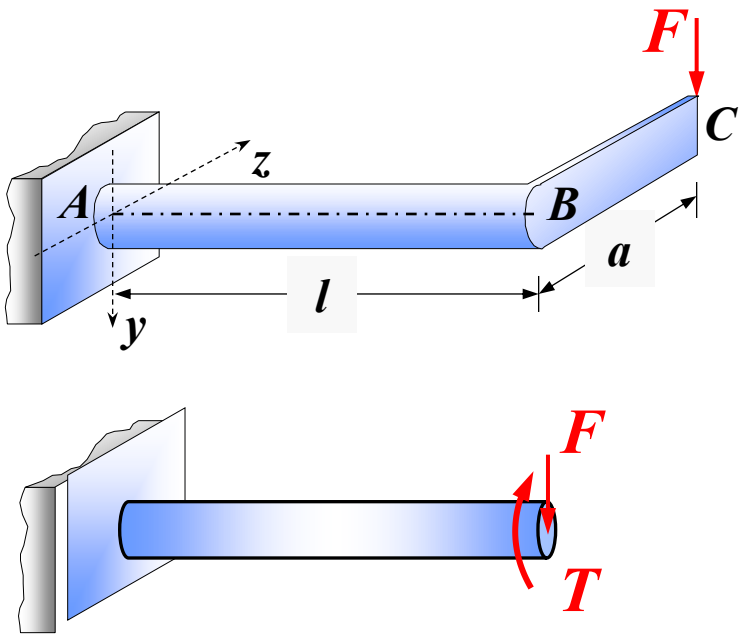
Tresca criterion: $\sqrt{\sigma^2 + 4\tau^2} \le [\sigma]$

von Mises criterion: $\sqrt{\sigma^2 + 3\tau^2} \le [\sigma]$

$$\Rightarrow \begin{cases} \text{Tresca criterion } \sqrt{(\frac{M_z}{W_z})^2 + 4(\frac{T}{W_P})^2} = \frac{1}{W_z}\sqrt{M_z^2 + T^2} \le [\sigma] \\ \text{von Mises } \sigma_{eff} = \sqrt{(\frac{M_z}{W_z})^2 + 3(\frac{T}{W_P})^2} = \frac{1}{W_z}\sqrt{M_z^2 + 0.75T^2} \le [\sigma] \end{cases}$$

Example 8.3

- For the circular shaft shown, $L = 50$ cm, $F = 8$ kN, $a = 37.5$ cm, $[\sigma] = 100$ MPa. Find the minimum diameter of shaft AB based on maximum shear stress criteria.



- Solution:

$$T = 3\ \text{kNm}, \quad M_{\max} = FL = 4\ \text{kNm}$$

Tresca criterion:

$$\sqrt{\sigma^2 + 4\tau^2} = \sqrt{\left(\frac{M_z}{W_z}\right)^2 + 4\left(\frac{T}{W_P}\right)^2}$$

$$= \frac{1}{W_z}\sqrt{M_z^2 + T^2} \le [\sigma]$$

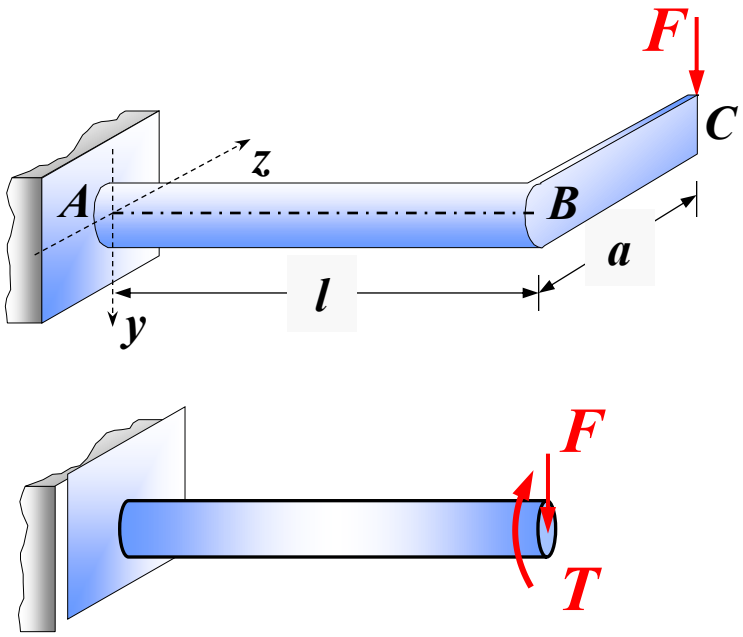
$$\Rightarrow W_z \ge \frac{\sqrt{M_z^2 + T^2}}{[\sigma]} = 5\times 10^{-5}\, m^3$$

$$\Rightarrow d \ge \sqrt[3]{\frac{32W_z}{\pi}} = 79.8mm$$

Question: What about the maximum distortion energy criteria?

Example 8.3

- For the circular shaft shown, $L = 50$ cm, $F = 8$ kN, $a = 37.5$ cm, $[\sigma] = 100$ MPa. Find the minimum diameter of shaft AB based on maximum distortion energy criteria.



$$T = 3\ \text{kNm}, \quad M_{\max} = FL = 4\ \text{kNm}$$

$$\sigma_{eff} = \sqrt{\sigma^2 + 3\tau^2} = \sqrt{\left(\frac{M_z}{W_z}\right)^2 + 3\left(\frac{T}{W_P}\right)^2}$$

$$= \frac{1}{W_z}\sqrt{M_z^2 + 0.75T^2} \le [\sigma]$$

$$\Rightarrow W_z \ge \frac{\sqrt{M_z^2 + 0.75T^2}}{[\sigma]} = 4.583\times10^{-5}\, m^3$$

$$\Rightarrow d \ge \sqrt[3]{\frac{32W_z}{\pi}} = 77.57mm$$